HEC (PHASE-II) PRACTICE TEST BY MCAT MCQs



INSTRUCTIONS:

- Duration of test is **150 MINUTES** (120 Questions).
- There is no any **NAGATIVE** marking.
- Fill only one circle for each question on given Response sheet. All questions carry equal marks.
- No mobile phones, calculators and wrist watches with calculators are allowed during test.
- After completion of test, please take a clear image of your response sheet and send it to our page by message.
- You can solve the practice test any time before displaying Answer Key.
- Answer Key will be displayed on our page MCAT MCQs (www.facebook.com/mcatmcqs) on **September 17, 2017 at 9:00 AM.**
- In the case of any confusion get help from admin.

BIOLOGY

Q.1 **They are the remains or impressions of living organisms preserved in rocks:**
A. Tissues
B. Fossils
C. Calcium
D. Statutes

Q.2 **The pushing out of materials by the cell against the concentration gradient is:**
A. Low transport
B. Passive transport
C. Active transport
D. Moderate transport

Q.3 **Which of the following organelles is present in both prokaryotic and eukaryotic cells?**
A. Chloroplast
B. Ribosomes
C. Mitochondria
D. Golgi Complex

Q.4 **Silk is chemically:**
A. Lipid
B. Wax
C. Protein
D. Carbohydrate

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- **Choose the word most similar in meaning to the capitalized ones:**

Q.51 **LOCAL**
A. Distant
B. Global
C. Nearby
D. Regional

Q.52 **OFFENSIVE**
A. Upsetting
B. Cruel
C. Aware
D. Normal

Q.53 **GRATEFUL**
A. Supportive
B. Thankful
C. Healthy
D. Kind

Q.54 **PERIL**
A. Doom
B. Happiness
C. Action
D. Danger

Q.55 **LEAGUE**
A. Apartment
B. Alliance
C. Individual
D. House

- **Choose the lettered word or phrase that is most nearly opposite in meaning to the word in capital letters:**

Q.56 **TYPICAL**
A. Hazardous
B. Average
C. Abnormal
D. Painful

Q.57 **HECTIC**
A. Careful
B. Calm
C. Alone
D. Specialized

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Q.36 **DNA polymerase is extracted from:**
A. Protists
B. Fungi
C. Virus
D. Bacteria

Q.37 **Organ functionally alike but structurally different are:**
A. Analogous
B. Homologous
C. Heterologous
D. Isologous

Q.38 **Succession end with a diverse and relatively stable:**
A. Xerosere
B. Climax community
C. Derosere
D. Hydrosere

Q.39 **Nutrients tends concentrate near bottom sediments in:**
A. Desert
B. Tundra
C. Aquatic ecosystem
D. Coniferous forest

Q.40 **As CFCs rise to the atmosphere, the ultraviolet rays release:**
A. Fluorine
B. Chlorine
C. Carbon
D. Hydrogen

ENGLISH

- **Complete the sentences by choosing the most appropriate option, from the given lettered choices (A to D) below each.**

Q.41 **He says he is a _____, but he can't play the piano or any other instrument and he can't sing.**
A. musician
B. magician
C. physician
D. technician

Q.42 **She hadn't eaten all day, and by the time she got home she was _____.**
A. blighted
B. confutation

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- **Question 58-60 are based on the following passage:**

But I did not want to shoot the elephant. I watched him beating his bunch of grass against his knees, with the *preoccupied grandmotherly* air that elephants have. It seemed to me that it would be murder to shoot him. I had never shot an elephant and never wanted to. (Somehow it always seems worse to kill large animal.) Besides, there was the beast's owner to be considered. But I had got to act quickly. I turned to some experienced-looking Burmans who had been there when we arrived, and asked them how the elephants had been behaving. They all said the same thing: he took no notice of you if you left him alone, but he might charge if you went too close to him.

Q.58 **The phrase 'Preoccupied grandmotherly air' signifies:**
A. Being totally unconcerned
B. Pretending to be very busy
C. A very superior attitude
D. Calm, dignified and affectionate disposition

Q.59 **From the passage it appears that the author was:**
A. An inexperienced hunter
B. Kind and considerate
C. Possessed with fear
D. A worried man

Q.60 **The author did not want to shoot the elephant because he:**
A. Was afraid of it
B. Did not have the experience of shooting big animals
C. Did not wish to kill animal which was not doing anybody any harm
D. Did not find the elephant to be ferocious

CHEMISTRY

Q.61 **Calculate the volume of O_2 at STP for the complete combustion of two moles of carbon disulphide which produce CO_2 and SO_2:**
A. 130 dm^3
B. 134.4 dm^3
C. 44.8 dm^3
D. 22.4 dm^3

Q.62 **Which of the following bonds have maximum bond energy?**
A. C—C
B. C=C
C. C≡C
D. N≡N

Q.63 **Molecular mass of water is:**
A. 15 amu

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B. 16 amu
C. 17 amu
D. 18 amu

Q.64 **Identify "X" in the reaction given below:**

$$CH_2{=}CH_2 + HBr \longrightarrow X$$

A. CH_2CH_2Br
B. CH_3CHBr
C. CH_3CH_2Br
D. CHCHBr

Q.65 **f-block elements from Ac to Lr are called:**
A. Lanthanides
B. Actinides
C. Outer transition
D. Both B & C

Q.66 **Surface tension of benzene at 20°C is:**
A. 72.583
B. 22.03
C. 28.88
D. 471.6

Q.67 **Alkali metals can be obtained in the free state by:**
A. Electrolysis of fused salts
B. Addition reaction
C. Oxidation
D. Solvay process

Q.68 **Which halogen molecule 'X_2' has lowest dissociation energy?**
A. F_2
B. Cl_2
C. Br_2
D. I_2

Q.69 **The anomalous electronic configuration shown by chromium and copper among 3-d series of elements is due to:**
A. Colour of ions of these metals
B. Stability associated with this configuration
C. Variable oxidation states of metals
D. Complex formation tendency of metals

Q.70 **Which one of the following is NOT a nucleophile?**
A. NH_2^-
B. BF_3
C. H_2O

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D. CH_3^+

Q.71 **Which one of the following test is given by both aldehyde and ketone?**
A. Silver mirror test
B. 2, 4 DNPH test
C. Fehling's solution test
D. Benedict's solution test

Q.72 **Borax exists in nature as:**
A. $Na_2B_4O_7.10H_2O$
B. $Na_2B_4O_7.5H_2O$
C. $Na_2B_4O_7.3H_2O$
D. $Na_2B_4O_7.H_2O$

Q.73 **The amount of energy released by absorbing an electron in the valence shell is:**
A. Ionization Energy
B. Electron Affinity
C. Atomic Radius
D. Atomization Energy

Q.74 **Rate of $=K[N_2O_5]$ has _____ of reaction:**
A. First order
B. Pseudo first order
C. Second order
D. Third order

Q.75 **The most dense element is:**
A. Li
B. K
C. Ca
D. Ba

Q.76 **Which one of the following molecules has shortest distance of carbon atoms?**
A. CH_3-CH_3
B. $CH_2=CH_2$
C. $CH\equiv CH$
D. $CH_2=CH_2-CH_3$

Q.77 **Which of the following elements has the lowest electronegativity?**
A. Cesium
B. Strontium
C. Calcium
D. Barium

Q.78 **Ketone reacts with Grignard's reagent in acidic medium forming:**

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A. Bond pairs of electrons
B. Unpaired electrons in atom or molecule
C. Lone pairs of electrons
D. Paired electrons in valence shells of electrons

Q.90 **The pH of solution is 5. To this solution sufficient acid was added and its pH decreases to 2. The increase in hydrogen ions concentration is:**
A. 10 times
B. 100 times
C. 1000 times
D. 2.5 times

PHYSICS

Q.91 **A nanosecond is:**
A. 10^{9} s
B. 10^{-9} s
C. 10^{-10} s
D. 10^{-08} s

Q.92 **The dimensional formula of angular velocity is:**
A. $M^{0}L^{0}T^{-1}$
B. MLT^{-1}
C. $M^{0}L^{0}T^{1}$
D. $ML^{0}T^{-2}$

Q.93 **A baseball is thrown vertically into the air. The acceleration of the ball at its highest point is:**
A. zero
B. g, down
C. g, up
D. 2g, down

Q.94 **The area under a velocity-time graph represents:**
A. Acceleration
B. Speed
C. Change in velocity
D. Displacement

Q.95 **An object is thrown straight up from ground level with a speed of 50 m/s. If g = 10 m/s2 its distance above ground level 6.0 s later is:**
A. 120 m
B. 270 m
C. 480 m
D. 0.00 m

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Q.82 What amount of NaOH is required to prepare 500 cm^3 of 0.1M solution:
A. 1g
B. 2g
C. 4g
D. 10g

Q.83 The substance used in the salt bridge of Galvanic cell is:
A. K_2SO_4
B. KCl
C. $K_2Cr_2O_7$
D. NaCl

Q.84 The oxygen atom in water is hybridized
A. Sp^3
B. Sp^2
C. Sp
D. dsp^2

Q.85 Starch is a polymer of
A. α-D-glucose
B. β-D-glucose
C. γ-D-glucose
D. α-L-glucose

Q.86 The increase in concentration of oxidizing agents in smog like H_2O_2, HNO_3, PAN and ozone in the air is called:
A. Carbonated smog
B. Nitrated smog
C. Sulphonated smog
D. Photochemical smog

Q.87 An alcohol is converted to an aldehyde with same number of carbon atoms as that of alcohol in the presence of $K_2Cr_2O_7/H_2SO_4$ the alcohol is:
A. $CH_3Cl(CH)_2OH$
B. $(CH_3)_3COH$
C. CH_3CH_2OH
D. $(CH_3)_2CHOH$

Q.88 1-chloropropane and 2-chlorpropane are isomers of each other, the type of isomerism in these two is called:
A. Cis-trans isomerism
B. Position isomerism
C. Chain isomerism
D. Functional group isomerism

Q.89 The paramagnetic character of a substance is due to:

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A. Ethyl alcohol
B. Butyl alcohol
C. Primary alcohol
D. Secondary alcohol

Q.79 **The maximum number of electrons in a shell with the principal quantum number equal to 4 is:**
A. 2
B. 10
C. 16
D. 32

Q.80 **Which of the following is used in manufacturing of Lozenges and Gargles?**
A. Esters
B. Phenols
C. Aldehydes
D. Ketones

Q.81 **Which one of the following formulae represents the organic compound formed when methylbenzene is heated under reflux with alkaline potassium manganate (VII) solution and the mixture then acidified?**

A.

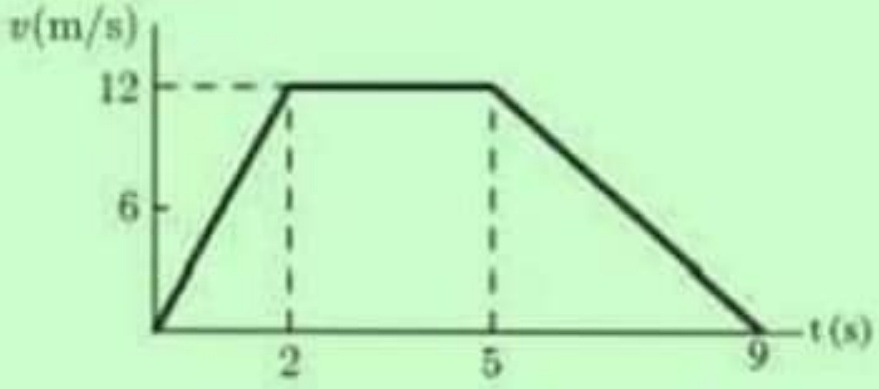

B.


C.


D.


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Q.96 The graph represents the straight line motion of a car. How far does the car travel between t = 2 s and t = 5 s?



A. 4 m
B. 12 m
C. 24 m
D. 36 m

Q.97 The inertia of a body tends to cause the body to:
A. Speed up
B. Slow down
C. Resist any change in its motion
D. Fall toward Earth

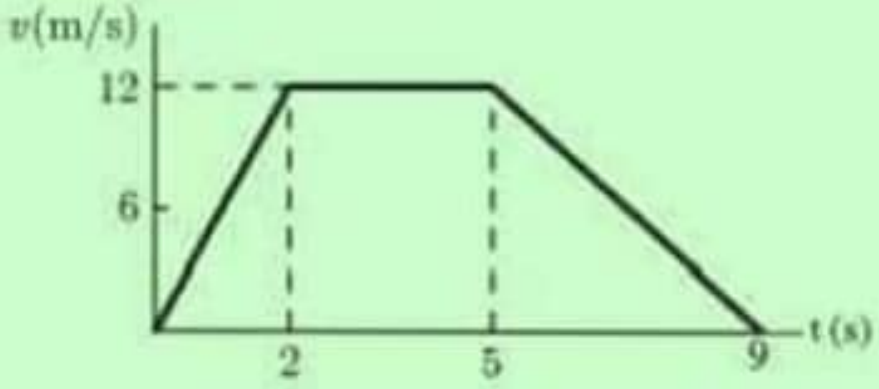
Q.98 A 6-kg object is moving south. A net force of 12N north on it results in the object having an acceleration of:
A. $2 m/s^2$, North
B. $2 m/s^2$, South
C. $6 m/s^2$, North
D. $6 m/s^2$, South

Q.99 A block slides down a frictionless plane that makes an angle of 30° with the horizontal. The acceleration of the block is:
A. $980 cm/s^2$
B. $566 cm/s^2$
C. $490 cm/s^2$
D. Zero

Q.100 An object moves in a circle. If the mass is tripled, the speed halved, and the radius unchanged, then the magnitude of the centripetal force must be multiplied by a factor of:
A. 3/2
B. 3/4
C. 9/4
D. 6

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A. 3/2
B. 3/4
C. 9/4
D. 6

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Q.101 A boy holds a 40-N weight at arm's length for 10 s. His arm is 1.5m above the ground. The work done by the force of the boy on the weight while he is holding it is:

A. 0 J
B. 6.1 J
C. 40 J
D. 60 J

Q.102 An elevator is rising at constant speed. Consider the following statements:

I. The upward cable force is constant.
II. The kinetic energy of the elevator is constant.
III. The gravitational potential energy of the Earth-elevator system is constant.
IV. The acceleration of the elevator is zero.
V. The mechanical energy of the Earth-elevator system is constant.

A. only II and V are true
B. only IV and V are true
C. only I, II, and III are true
D. only I, II, and IV are true

Q.103 Two bodies, A and B, have equal kinetic energies. The mass of A is nine times that of B. The ratio of the momentum of A to that of B is:

A. 1:9
B. 1:3
C. 1:1
D. 3:1

Q.104 One revolution is the same as:

A. 57 rad
B. $\pi/2$ rad
C. π rad
D. 2π rad

Q.105 An object attached to one end of a spring makes 20 vibrations in 10 s. Its frequency is:

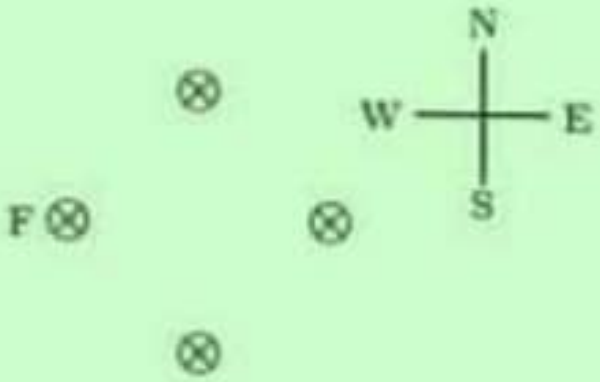
A. 2Hz
B. 10 s
C. 0.05 Hz
D. 2 s

Q.106 A column of argon is open at one end and closed at the other. The shortest length of such a column that will resonate with a 200 Hz tuning fork is 42.5 cm. The speed of sound in argon must be:

A. 85.0m/s
B. 170m/s
C. 340m/s
D. 470m/s

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Q.113 **Four long straight wires carry equal currents into the page as shown. The magnetic force exerted on wire F is:**



A. North
B. East
C. South
D. West

Q.114 **The relation "n₁ Sin θ₁ = n₂ Sin θ₂", which applies as a ray of light strikes an interface between two media, is known as:**
A. Gauss' law
B. Snell's law
C. Faraday's law
D. Cole's law

Q.115 **Starting with a sample of pure ^{64}Cu, 7/8 of it decays into Zn in 15 minutes. The corresponding half-life is:**
A. 15 minutes
B. 5 minutes
C. 7 minutes
D. 3.75 minutes

Q.116 **Which one of the following is NOT needed in a nuclear fission reactor?**
A. Moderator
B. Coolant
C. Control device
D. Accelerator

Q.117 **In Compton scattering from stationary particles the maximum change in wavelength can be made smaller by using:**
A. Higher frequency radiation
B. Lower frequency radiation
C. More massive particles
D. Less massive particles

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Q.118 **Light from some stars shows an apparent change in frequency because of:**

A. Interference
B. Diffraction
C. Reflection
D. Relative motion

Q.119 **In connection with X-ray emission the symbol Lβ refers to:**

A. Beta particle radiation
B. An atomic state of angular momentum h/2π
C. The inductance associated with an orbiting electron
D. X-radiation associated with an electron going from n = 4 to n = 2

Q.120 **In a Young's double-slit experiment, the slit separation is doubled. To maintain the same fringe spacing on the screen, the screen-to-slit distance "D" must be changed to:**

A. D/2
B. $D/\sqrt{2}$
C. $D\sqrt{2}$
D. 2D

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C. ravenous
D. ostentatious

Q.43 **Steven is always _____ about showing up for work because he feels that tardiness is a sign of irresponsibility.**
A. legible
B. literal
C. tolerable
D. punctual

Q.44 **It usually _____ here in June:**
A. Rains
B. Rained
C. Will rain
D. Rain

Q.45 **I would like to have this suit_____:**
A. Clean
B. Cleaning
C. Cleaned
D. To clean

Q.46 **Let us go to the bookshop as soon school_____:**
A. Will be over
B. Was over
C. Be over
D. Is over

- **Identify the word or phrase that needs to be changed for the sentence to be correct:**

Q.47 I go by the post office every morning on my way too work.
A B C D

Q.48 She has been studying since 7 hours. No wonder why her teachers praise her hard work all
A B C D
the time.

Q.49 I was surprised to hear that it was the seven-hour trip up green mountains which they
A B
completed so fast.
C D

Q.50 The sentiments in the common people must be kept in mind when the Government decides to
A B C D
pass the Bill.

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HAT-UG-M (MCAT MCQs)

BOOK COLOR GREEN

TEST TIME 9:00AM (September 16, 2017)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q.1	B	Q.31	C	Q.61	B	Q.91	B
Q.2	C	Q.32	A	Q.62	D	Q.92	A
Q.3	B	Q.33	C	Q.63	D	Q.93	B
Q.4	C	Q.34	A	Q.64	C	Q.94	D
Q.5	C	Q.35	D	Q.65	B	Q.95	A
Q.6	C	Q.36	D	Q.66	C	Q.96	D
Q.7	B	Q.37	A	Q.67	A	Q.97	C
Q.8	B	Q.38	B	Q.68	A	Q.98	A
Q.9	B	Q.39	C	Q.69	B	Q.99	C
Q.10	B	Q.40	B	Q.70	B	Q.100	B
Q.11	B	Q.41	A	Q.71	B	Q.101	A
Q.12	D	Q.42	C	Q.72	A	Q.102	D
Q.13	D	Q.43	D	Q.73	B	Q.103	D
Q.14	B	Q.44	A	Q.74	A	Q.104	D
Q.15	B	Q.45	C	Q.75	D	Q.105	A
Q.16	B	Q.46	D	Q.76	C	Q.106	C
Q.17	B	Q.47	D	Q.77	A	Q.107	B
Q.18	B	Q.48	B	Q.78	B	Q.108	D
Q.19	B	Q.49	B	Q.79	D	Q.109	C
Q.20	B	Q.50	B	Q.80	B	Q.110	A
Q.21	B	Q.51	C	Q.81	B	Q.111	B
Q.22	A	Q.52	A	Q.82	B	Q.112	D
Q.23	D	Q.53	B	Q.83	B	Q.113	B
Q.24	D	Q.54	D	Q.84	A	Q.114	B
Q.25	B	Q.55	B	Q.85	A	Q.115	B
Q.26	C	Q.56	C	Q.86	D	Q.116	D
Q.27	D	Q.57	B	Q.87	C	Q.117	C
Q.28	C	Q.58	D	Q.88	B	Q.118	D
Q.29	B	Q.59	B	Q.89	B	Q.119	D
Q.30	D	Q.60	B	Q.90	C	Q.120	D

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